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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: **TEEG104/TEUP104/TEJS104**

COURSE TITLE: **LEARNING, TEACHING AND APPLYING GEOMETRY AND HANDLING**

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

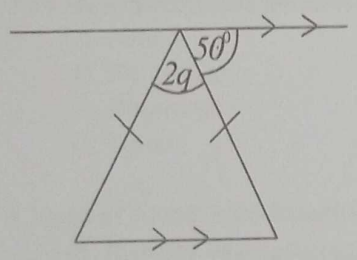
SECTION A

[25 marks]

Answer all the questions in this section.

1. The government of Ghana decided to build a chip compound for three compounding villages A, B and C. An architect was engaged to help locate a place for the project, so he located the project at a point which is equidistant from the paths connecting the three villages. This means that the architect bisected the _____ connecting the villages.
   A. paths
   B. angles
   C. centre
   D. squares

2. Find the value of q in the diagram below:



   A. 25°
   B. 40°
   C. 50°
   D. 80 °

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3. The concept of alternate and corresponding angles is very important in Geometry. Why are they called corresponding angles?
   A. They lie below the transversal of given parallel lines.
   B. They lie above the transversal of given parallel lines.
   C. They lie in the same position with the transversal.
   D. They are formed within the given points of the transversal.

4. The concepts of congruency, similarity and symmetry can be understood from the perspective of geometric transformation. Which of the transformations result in the formation of congruent shapes?
   A. Reflection and enlargement
   B. Reflection and rotation
   C. Reflection and translation
   D. Reflection and reduction

5. Which of the following geometrical concepts is zero dimensional and non-directional?
   A. Line
   B. Line segment
   C. Point
   D. Ray

6. A JHS 1 pupil was asked to identify an angle a such that $90^o < a < 180^o$. Which of the following will be the correct response?
   A. Acute angle
   B. Obtuse angle
   C. Reflect angle
   D. Straight line angle

7. Which one of the following properties of angles formed by a transversal on parallel lines is NOT true?
   A. Adjacent angles are equal.
   B. Alternate angles are equal.
   C. Correspondent angles are equal.
   D. Vertically opposite angles are equal.

8. Given that $tanx = \frac{5}{12}$, where $0 < x < 90°$, what is the value of $sinx + cosx$?

   A. $\frac{17}{13}$
   B. $\frac{125}{156}$
   C. $\frac{209}{156}$
   D. $\frac{13}{17}$

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9. Two complementary angles are in the ratio 2:3. Find the difference between the angles.
   A. 18°
   B. 36°
   C. 54°
   D. 90°
10. The average score of 4 students in a science experiment is 72. If a fifth student joins the group and the new mean becomes 74, what is the score of the fifth student?
   A. 80
   B. 82
   C. 84
   D. 88

11. Which of the following sets of numbers will NOT constitute the dimensions of any triangle?
   A. $5m, 8m$ and $7m$
   B. $6m, 8m$, and $10m$
   C. $7m, 7m$ and $7m$
   D. $5m, 7m$, and $13m$

12. A polygon has 12 sides. How many triangles can be found in this polygon?
   A. 10
   B. 12
   C. 150
   D. 1800

13. In trigonometry, the circle is divided into four quadrants, which is measured from the anticlockwise direction. Which trigonometric ratio is positive in the 3rd quadrant?
   A. Sine of the angle
   B. Cosine of the angle
   C. Tangent of the angle
   D. Cotangent of the angle

14. Mrs. Halima, a class four teacher asked her pupils to identify a quadrilateral whose unequal diagonals do not only bisect each other but are also perpendicular to each other. What answer does Mrs. Halima expects from the pupils?
   A. Kite
   B. Rectangle
   C. Rhombus
   D. Square

15. A locus of points equidistant from a fixed point is a _____.
   A. bisector of an angle.
   B. circle with centre at a fixed point.
   C. perpendicular bisector of a given line.
   D. bisection of two sides or three sides of a triangle.

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16. Which of the following is NOT a Pythagorean triplet?
    A. (5, 12, 13)
    B. (7, 24, 25)
    C. (7, 40, 41)
    D. (9, 60, 61)

17. A ladder 13 m long is placed on the ground in such a way that it touches the top of a vertical wall 12 m high. Find the distance of the foot of the ladder from the bottom of the wall.
    A. 5m
    B. 25m
    C. 313m
    D. $\sqrt{313}m$

18. When an aeroplane is 800m above the ground, the angle of elevation from a point P on the ground is $30°$. How far is the plane from P?
    A. 1385m
    B. 692.8m
    C. 1600m
    D. 356.6m

19. The diagonal of a rectangle is 17cm. if the length of the rectangle is 15cm, calculate the area of the rectangle.
    A. $8cm^2$
    B. $120cm^2$
    C. $136cm^2$
    D. $255cm^2$

20. If the vector $\begin{pmatrix} x \\ 8 \end{pmatrix}$ is perpendicular to the vector $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ find the value of x.
    A. -5
    B. 5
    C. 14
    D. -14

21. If $\overline{PQ} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ and $\overline{PR} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ find $\overline{RQ}$
    A. $\begin{pmatrix} -8 \\ 12 \end{pmatrix}$
    B. $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$
    C. $\begin{pmatrix} 4 \\ -4 \end{pmatrix}$
    D. $\begin{pmatrix} -4 \\ 4 \end{pmatrix}$

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22. The bearing of town A from B is 125°. Determine the bearing to B from A.
    A. 35°
    B. 55°
    C. 125°
    D. 305°

23. The parallel sides of a trapezium are 11cm and 8cm. If the distance between the parallel lines is 9cm, find the area of the trapezium.
    A. $28cm^2$
    B. $85.5cm^2$
    C. $792cm^2$
    D. $396cm^2$

24. Scores in Mathematics test have a mean score of 49 and a standard deviation of 4. What is the variance of the scores?
    A. 2
    B. 7
    C. 16
    D. 196

25. A bag contains 3 white and 4 black identical balls and another bag contains 4 white and 2 black identical balls. A ball is selected at random from each bag. Find the probability that the two balls selected are of the same colour.
    A. $\frac{1}{21}$
    B. $\frac{4}{21}$
    C. $\frac{10}{21}$
    D. $\frac{11}{21}$

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SECTION B

[75 marks]

Answer three questions from this section.

1a. A regular polygon has its exterior angle 100^0 less than the interior angle. Find, **10Marks**

i) the number of sides.
ii) the sum of its interior angles.

1b. Describe an activity you would use to assist a JHS 2 student to deduce that the sum of the interior angles of a polygon is given as given $180(n-2)$. **10Marks**

1c. A parallelogram ABCD has vertices A(1,3), B(-2,5), C(1,8) and D(m,n), find the coordinates of D. **5Marks**

2a. P, Q and R are three ships at sea. The bearing of Q from P is 310^0 and the bearing R from P is 220^0. If $|PR| = 13km$ and $|PQ| = 12km$, calculate **10Marks**

i) $|QR|$
ii) the bearing of R from Q.

2b. An open cylindrical tin has a height of $60mm$ and a diameter of $154mm$. Calculate the **10marks**

i) area of the cross-section.
ii) volume of the cylinder.
iii) curved surface area.

The total surface area. (take $\pi = \frac{22}{7}$)

2c. Classify the following as either nominal data or discrete data: **5Marks**

The number of students in a mathematics class.
The colours of jerseys of a football team.
The number of jerseys of a football team.
The nationality of level 100 students in Early Grade.
The number of goals scored in a game.

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3. (a) i . Construct triangle ABC such that $|AB| = 8cm$, $\angle ABC = 30^0$ and $\angle BAC = 105^0$ **25Marks**

ii. The locus l_1 of points equidistant from A and B

iii. The locus l_2 of points equidistant from B and C

(b). Locate P, the point of intersection of l_1 and l_2.

(c). Using PC as radius draw a circle

(d). Measure

i. $|BC|$

ii. The radius of the circle.

4a. Using diagrams, indicate the following towns on a globe **10Marks**

i) A $(40^0S, 60^0W)$

ii) M $(50^0N, 40^0E$

iii) T $(60^0S, 20^0E$

iv) B $(20^0N, 30^0W$

v) Q $(30^0S, 70^0E$

4b. The vectors $p = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, q = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $r = \frac{1}{2}(q - p)$ **10Marks**

i) Find vector r

ii) If $mp + nq = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$, find the values of m and n, where m and n are scalars.

4c. The length of the shadow of the administration block, 89m tall, is 93m. Find the angle of elevation of the top of the building. **5marks**

5a. The table below gives the distribution of marks by 30 level 100 students in a quiz. **13Marks**

Marks	2	3	4	5	6	7	8
Frequency	3	5	4	7	5	4	2

i) Find the mean of the distribution

ii) Find the Median

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iii) What is the probability of selecting a student who scores at least 7 marks?

5b. A number is chosen at random from the set $U = \{6, 9, 13, 14, 16, 19, 21, 26\}$. What is the probability that the number is: **8Marks**

i) odd.

ii) prime.

iv) multiple of 3.

iv) greater than 20.

5c. Differentiate between qualitative data and quantitative data. **4marks**